

Bidirectional Vietnamese Dialect Transfer

Region-Conditioned Standard→Dialect Generation
with Multi-Task Seq2Seq and Sampling Back-Translation

Capstone — IT4772E Natural Language Processing

Hanoi University of Science and Technology

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The problem

- Vietnamese has three dialect regions: **Northern**, **Central**, **Southern**.
- Example (Central): “Anh đi **mô** rồi?” = standard “Anh đi **đâu** rồi?” (“Where did he go?”).
- **All prior work normalizes** dialect → standard (ViDia2Std, NADI for Arabic).
- **Nobody generates the reverse**: standard → dialect, conditioned on a region.

Our goal

One model, both directions *and* three regions — normalization **and** the novel region-conditioned dialect *generation*.

- ① **Novel task**: region-conditioned standard→dialect generation (Feature C) for all three regions.
- ② **Unified multi-task model**: one BARTpho-syllable does (A) dialect→std, (B) lexical normalization, (C) std→dialect — selected by a text task prefix.
- ③ **Evaluation suite for one-to-many**: Dialect Feature Recall, marker-independent Dialectal Edit-Recall, region accuracy, copy-rate (BLEU alone is misleading here).
- ④ **Data strategy**: oversampling + sampling back-translation for the data-starved regions.

Why is generation harder than normalization?

Dialect $\rightarrow$ Standard (A)

- Many-to-one (converging)
- One correct answer
- BLEU is meaningful

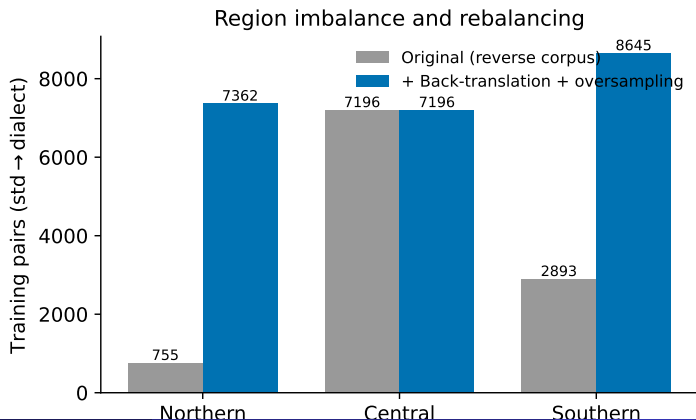
Standard $\rightarrow$ Dialect (C)

- **One-to-many** (diverging)
- Many valid realizations
- **Copying scores high BLEU** but does nothing

Consequence

We need metrics that ask *“was the right kind of change made?”*, not *“does it match one reference?”*

- **ViDia2Std** (Ta et al., AAAI-26): 13,657 dialect/standard pairs, 63 provinces, region-labelled. Used forward (A) *and* reversed (C).
- **ViLexNorm** (EACL 2024): 10,467 teencode→standard pairs (Feature B).
- **Severe imbalance**: Central $\approx 66\%$, Southern $\approx 27\%$, Northern $\approx 7\%$.

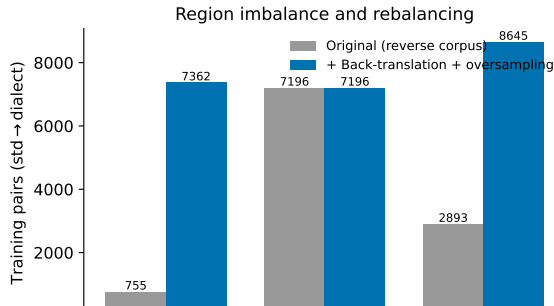


Method (1): multi-task task-prefix model

- Input string: $\{\text{task}\}: \{\text{source}\} \rightarrow$ one model, no new heads.
- $P_{\theta}(y \mid x, c) = \prod_t P_{\theta}(y_t \mid y_{<t}, x, c)$, $c = \text{task/region prefix}$.
- **BARTpho-syllable**: syllable tokenization is robust to dialect/teencode OOV (no fragile word segmenter).
- Prefixes: `dialect2std`, `std2dialect_{northern,central,southern}`, `lexnorm`.

Method (2): reverse corpus + sampling back-translation

- **Reverse corpus**: swap (dialect, standard) $\rightarrow$ (standard, dialect) + region prefix. Target side stays human-authored.
- **Sampling back-translation** (Edunov et al. 2018): keep *real dialect* as target, *sample* synthetic standard sources from the strong A-model. Sampling, not beam, to avoid mode collapse.
- **Rebalancing**: oversample Northern/Southern. After augmentation the three regions are roughly balanced.



Method (3): transfer-aware model selection

Problem

Pooled validation BLEU rewards a *copying* checkpoint (copying scores high on overlap for std→dialect).

Fix

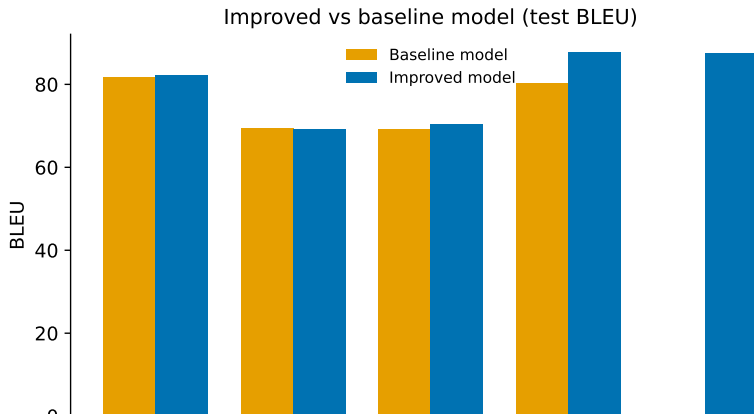
Select on a **macro per-task BLEU + Dialectal Edit-Recall** composite, so a checkpoint is preferred only if it actually performs the dialectal substitutions.

- **Overlap**: BLEU, chrF, ROUGE-L, METEOR; **semantic**: BERTScore; **normalization**: WER/CER (Feature A), ERR (Feature B).
- **Dialect Feature Recall (DFR)**: fraction of the reference's regional markers reproduced.
- **Dialectal Edit-Recall** (marker-independent): does the output make the same token substitutions the gold makes? (Copy $\Rightarrow$ Edit-Recall = 0.)
- **Region accuracy + copy-rate**: controllability and under-generation.

Results: normalization (Feature A)

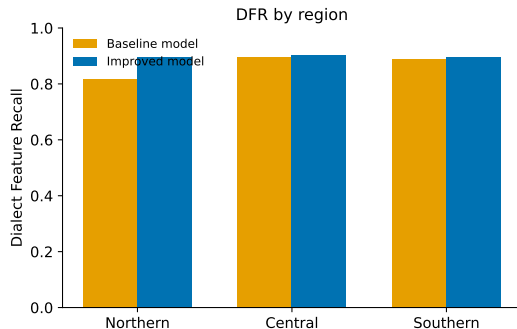
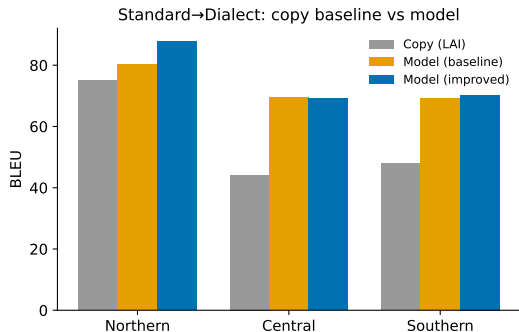
- Our BARTpho-syllable (396M): BLEU **82.10**, WER **0.112**, CER **0.069**.
- ViDia2Std best (mBART-large-50, 611M): BLEU 81.66, WER 0.123, CER 0.075.
- ViDia2Std BARTpho-syllable-*base* (132M): BLEU 75.97.

⇒ **Matches and edges the larger SOTA model with fewer parameters.**



Results: dialect generation (Feature C)

- Improved model overall DFR ≈ 0.90 , Edit-Recall **0.64–0.71**, copy-rate **0.005–0.090** (far below the copy floor of 1.0).
- Big gains on **Northern** (data-starved): copy-rate **0.209 $\rightarrow$ 0.090**, Edit-Recall **0.385 $\rightarrow$ 0.635**, BLEU **80.2 $\rightarrow$ 87.7**, DFR **0.817 $\rightarrow$ 0.897**.



- **Under-transfer / copying** (copy-rate, Edit-Recall) — worst on Northern.
- **Wrong-region leakage** (region accuracy) — markers of another region appear.
- **Over-transfer** (Edit-Precision) — spurious markers.
- **Semantic drift** (BERTScore / human eval).
- One-to-many **diversity**: sampling yields varied valid realizations (Self-BLEU: Central 81.6, Southern 86.7, Northern 93.7; lower = more diverse).

Limitations & future work

- Single dialectal reference per sentence; 63 provinces collapsed to 3 regions.
- DFR depends on a marker list (mitigated by marker-independent Edit-Recall).
- Northern remains smallest; back-translation sources are synthetic.
- Future: human evaluation at scale, ViT5 comparison, iterative back-translation rounds.

- First **region-conditioned standard→dialect** generator for Vietnamese.
- One multi-task model: normalization *and* generation *and* lexical normalization.
- Normalization competitive with a much larger SOTA model.
- New transfer-aware metrics expose what BLEU hides; data strategy fixes regional imbalance.

Thank you!

Questions?